WORKED SOLUTION

STEP III 2015, Question 8

Pure • Polar Differential Equations

STEP III 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 8.4, *Spiralling onto a Circle*: https://beyondstep.keithmcnulty.org/ch08/limit-cycle.html

The key idea

Both differential equations are built from the combinations $x\,\mathrm{d}y - y\,\mathrm{d}x$ and $x\,\mathrm{d}x + y\,\mathrm{d}y$, and these are exactly what polar coordinates simplify: they become $r^2\,\mathrm{d}\theta$ and $r\,\mathrm{d}r$. So we treat θ as the parameter along the curve, write $\mathrm{d}y/\mathrm{d}x$ as a ratio of θ-derivatives, and let the trigonometry collapse. In part (ii) the extra terms turn the equation into a Bernoulli equation for r, which the substitution $u = r^{-2}$ makes linear.

Solution

Write r' for $\mathrm{d}r/\mathrm{d}\theta$, $c = \cos\theta$ and $s = \sin\theta$. Along a curve $r = r(\theta)$,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = r'c - rs, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = r's + rc, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta}.$$

Multiplying either equation through by $\mathrm{d}x/\mathrm{d}\theta$, an equation of the form $P\dfrac{\mathrm{d}y}{\mathrm{d}x} = Q$ becomes

$$P\frac{\mathrm{d}y}{\mathrm{d}\theta} - Q\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0.$$

Two identities do all the work:

$$x\frac{\mathrm{d}y}{\mathrm{d}\theta} - y\frac{\mathrm{d}x}{\mathrm{d}\theta} = rc(r's + rc) - rs(r'c - rs) = r^2(c^2 + s^2) = r^2,$$
$$x\frac{\mathrm{d}x}{\mathrm{d}\theta} + y\frac{\mathrm{d}y}{\mathrm{d}\theta} = rc(r'c - rs) + rs(r's + rc) = rr'.$$

Part (i)

Here $P = y + x$ and $Q = y - x$, so

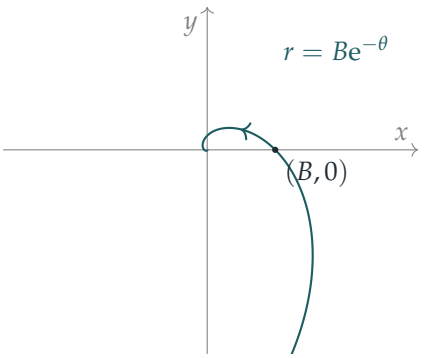
$$(y+x)\frac{\mathrm{d}y}{\mathrm{d}\theta} - (y-x)\frac{\mathrm{d}x}{\mathrm{d}\theta} = \left(x\frac{\mathrm{d}y}{\mathrm{d}\theta} - y\frac{\mathrm{d}x}{\mathrm{d}\theta}\right) + \left(x\frac{\mathrm{d}x}{\mathrm{d}\theta} + y\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)$$
$$= r^2 + rr'.$$

So the equation is $r(r' + r) = 0$, and since $r > 0$ we may divide by r to get

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} + r = 0,$$

as required.

This is separable (or just recognise exponential decay): $r = B\mathrm{e}^{-\theta}$ with $B > 0$. This is an *equiangular* (logarithmic) spiral. As θ increases the curve winds anticlockwise into the origin, never reaching it; as θ decreases it winds outwards without bound. Since $r/r' = -1$, the curve cuts every ray from the origin at the same angle, 45° to the inward radius. Different B give rotations of the same spiral, because $B\mathrm{e}^{-\theta} = \mathrm{e}^{-(\theta - \ln B)}$.



The arrow shows the direction of increasing θ. The spiral shrinks by a factor $\mathrm{e}^{2\pi} \approx 535$ on each turn, so only about one turn is visible at any scale; the rest is packed tightly around the origin.

Part (ii)

Now $P = y + x - x(x^2 + y^2)$ and $Q = y - x - y(x^2 + y^2)$, with $x^2 + y^2 = r^2$. The new terms contribute

$$-r^2\left(x\frac{\mathrm{d}y}{\mathrm{d}\theta} - y\frac{\mathrm{d}x}{\mathrm{d}\theta}\right) = -r^2 \cdot r^2 = -r^4,$$

so, using part (i) for the old terms, the equation becomes

$$r^2 + rr' - r^4 = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}r}{\mathrm{d}\theta} = r^3 - r,$$

again dividing by $r > 0$.

Ask yourself How do we deal with the r^3 term, which stops the equation being linear?

This is a Bernoulli equation. The target answer is in terms of r^2, which suggests working with $u = r^{-2}$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -2r^{-3}\frac{\mathrm{d}r}{\mathrm{d}\theta} = -2r^{-3}(r^3 - r) = -2 + 2u.$$

So $u' - 2u = -2$, a linear equation with constant coefficients: $u = 1$ is a particular solution and $A\mathrm{e}^{2\theta}$ solves the homogeneous equation, giving

$$\frac{1}{r^2} = 1 + A\mathrm{e}^{2\theta} \qquad \Longrightarrow \qquad r^2 = \frac{1}{1 + A\mathrm{e}^{2\theta}},$$

which is the required form. (Separating variables and using partial fractions on $\dfrac{1}{r(r-1)(r+1)}$ gives the same result with more effort.) The curve exists only where $1 + A\mathrm{e}^{2\theta} > 0$.

The shapes of the solutions

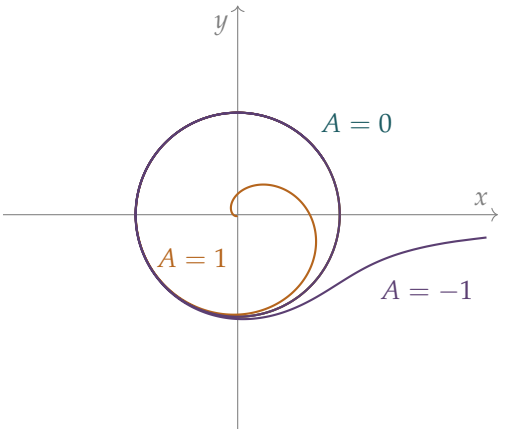
First note that if $A = \pm\mathrm{e}^{-2\alpha}$ then $A\mathrm{e}^{2\theta} = \pm\mathrm{e}^{2(\theta-\alpha)}$, so the curve for A is the curve for $A = \pm 1$ rotated through α. So there are really only three shapes, according to the sign of A.

- $A = 0$: $r = 1$, the unit circle.
- $A > 0$: $1 + A\mathrm{e}^{2\theta} > 1$, so $r < 1$ and the curve is defined for every θ. Since $r' = r(r^2 - 1) < 0$, r decreases steadily. As $\theta \to -\infty$, $r \to 1$, so the curve winds out towards the unit circle from inside; as $\theta \to +\infty$, $r \approx A^{-1/2}\mathrm{e}^{-\theta} \to 0$, so it spirals into the origin, looking more and more like the spiral of part (i).
- $A < 0$: we need $\mathrm{e}^{2\theta} < -1/A$, that is $\theta < \theta_0 = \frac{1}{2}\ln(-1/A)$. Now $r > 1$ and $r' > 0$. As $\theta \to -\infty$, $r \to 1$ from above, so the curve winds in towards the unit circle from outside; as $\theta \to \theta_0^-$, $r \to \infty$. Near θ_0, $1 + A\mathrm{e}^{2\theta} \approx 2(\theta_0 - \theta)$ (since $A\mathrm{e}^{2\theta_0} = -1$), so $r \approx \big(2(\theta_0 - \theta)\big)^{-1/2}$, and the distance from the curve to the line $\theta = \theta_0$ is

  $$r\sin(\theta_0 - \theta) \approx r(\theta_0 - \theta) \approx \frac{1}{2r} \to 0.$$

  So the curve goes off to infinity with the line through the origin at angle θ_0 as an asymptote, approaching it from the clockwise side. For $A = -1$, $\theta_0 = 0$ and the curve ends up close to $y = -\frac{1}{2x}$ for large x.

The unit circle is a limit cycle: every other solution spirals towards it as $\theta \to -\infty$, and away from it as θ increases.



The plum curve ($A = -1$) runs off to infinity along the positive x-axis, just below it.

Checks

- Substituting $r = Be^{-\theta}$ back into the original equation of (i) at a few points (convert to x, y and $\mathrm{d}y/\mathrm{d}x$) gives zero; likewise for the formula of (ii) with several values of A of both signs.
- $A = 0$ gives $r = 1$, and indeed $r' = r^3 - r$ is satisfied by the constant $r = 1$; in x, y, the circle $x^2 + y^2 = 1$ makes the extra terms cancel the old ones exactly.
- Starting from $r(0) = \frac{1}{2}$ (so $A = 3$), the formula gives $r(2) = 1/\sqrt{1 + 3e^4} \approx 0.0779$, which agrees with a step-by-step numerical solution of $r' = r^3 - r$ to many decimal places.
- For $A = -1$ at $\theta = -10^{-4}$, $r \approx 70.71$ and the distance to the x-axis is $0.00707 \approx 1/(2r)$, confirming the asymptote.

Common slips

- Forgetting that $\mathrm{d}y/\mathrm{d}x$ must be found as $(\mathrm{d}y/\mathrm{d}\theta)/(\mathrm{d}x/\mathrm{d}\theta)$ with r varying, i.e. dropping the r' terms.
- Missing the restriction $1 + Ae^{2\theta} > 0$ when $A < 0$, and so not noticing that these curves go off to infinity at a finite angle.
- Sketching only one curve for each sign of A without saying that the others are rotations of it.

Beyond STEP

The equation $r' = r^3 - r$ is a polar picture of a planar dynamical system, and the unit circle is a *limit cycle*: an isolated closed orbit that nearby trajectories spiral towards or

away from. Systems of this kind, where the radial equation decouples, are the standard first examples in courses on nonlinear differential equations and phase-plane analysis, leading to the Poincaré–Bendixson theorem. Boyce and DiPrima, *Elementary Differential Equations*, has a good chapter on nonlinear systems and limit cycles.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.